



Geometric Series

The sum of the terms of an arithmetic sequence is an arithmetic series. Likewise, you can sum the terms of a geometric sequence to get a geometric series.

Definition

A **geometric series** is a series that results from adding the terms of a geometric sequence.

For example, if you start with the geometric sequence

$$2, -6, 18, -54, 162, \dots,$$

then you can define the corresponding series.

$$2 + (-6) + 18 + (-54) + 162 + \dots$$

► **Remember** The iterative rule for a geometric sequence is

$$a_n = a_1 \cdot r^{n-1}.$$

Finding the Sum of the First n Terms of a Geometric Series

A general formula for the sum, S_n , of the first n terms of a geometric series can be seen with the help of an example.

Consider a geometric series where $a_1 = 1$ and $r = 2$. Now consider the 8th partial sum.

$$S_8 = 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128$$

Multiply both sides of the equation by 2, then subtract.

$$\begin{array}{r} S_8 = 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 \\ - 2S_8 = - (2 + 4 + 8 + 16 + 32 + 64 + 128 + 256) \\ \hline (1-2)S_8 = 1 + 0 + 0 + 0 + 0 + 0 + 0 + 0 - 256 \end{array}$$

$$S_8 = \frac{1 - 256}{(1 - 2)}$$

In the numerator, notice that $1 = a_1$ and that $256 = a_9$. In the denominator, $2 = r$.

You can find the general rule for the sum of the first n terms of a geometric series as follows:

$$\begin{array}{r} S_n = a_1 + a_1r + a_1r^2 + \dots + a_1r^{n-2} + a_1r^{n-1} \\ - rS_n = - (a_1r + a_1r^2 + \dots + a_1r^{n-2} + a_1r^{n-1} + a_1r^n) \\ \hline S_n - rS_n = a_1 - a_1r^n \\ S_n(1 - r) = a_1 - a_1r^n \\ S_n = \frac{a_1 - a_1r^n}{1 - r} \\ S_n = \frac{a_1(1 - r^n)}{1 - r}, \text{ where } r \neq 1 \end{array}$$

Geometric Series Formula

The sum of the first n terms of a geometric series with first term a_1 and common ratio r is

$$S_n = \frac{a_1(1 - r^n)}{1 - r}$$

where $r \neq 1$.

► **Think About It** Dividing by zero is undefined, which is why $r \neq 1$ in the formula for the sum of the first n terms of a geometric series.

EXAMPLE 1

Find the indicated sum of the geometric series.

A S_7 for the series $400 + 200 + 100 + 50 + \dots$

SOLUTION

Notice that $n = 7$, $a_1 = 400$, and $r = \frac{1}{2}$.

$$S_7 = \frac{400 \left(1 - \left(\frac{1}{2} \right)^7 \right)}{1 - \frac{1}{2}} = \frac{400 \left(1 - \frac{1}{128} \right)}{\frac{1}{2}} = \frac{400 \cdot \frac{127}{128}}{\frac{1}{2}} = 793\frac{3}{4}$$

The sum is $793\frac{3}{4}$.

B $\sum_{i=2}^6 3^i$

SOLUTION

Here, $a_1 = 9$, $n = 6$ and $r = 3$.

$$S_6 = \frac{9(1 - 3^6)}{1 - 3} = \frac{9(1 - 729)}{-2} = \frac{-6552}{-2} = 3276$$

The sum is 3276. ■

An Alternate Formula for the Sum of the First n Terms of a Geometric Series

You can see that, to use the formula for a geometric series, you need to know the first term, the common ratio, and the number of terms being summed.

With a little work, you can also derive a formula for the n th partial sum of a geometric series when you know the first term, the common ratio, and the value of the n th term.

Multiply both sides of the iterative rule for a geometric sequence by r .

$$\begin{aligned}a_n &= a_1 r^{n-1} \\a_n \cdot r &= a_1 r^{n-1} \cdot r \\a_n r &= a_1 r^{n-1+1} \\a_n r &= a_1 r^n\end{aligned}$$

Replacing $a_1 r^n$ with $a_n r$ in the formula $S_n = \frac{a_1 - a_1 r^n}{1 - r}$ yields the alternate geometric series formula.

Alternate Geometric Series Formula

The sum of the first n terms of a geometric series is given by

$$S_n = \frac{a_1 - a_n r}{1 - r}$$

where $r \neq 1$.

► **Think About It** This formula does not require you to know n .

EXAMPLE 2

Find the sum of the geometric series.

A $3 - 6 + 12 - 24 + \dots + 192$

SOLUTION

Notice that $a_1 = 3$, $r = -2$, and $a_n = 192$. Use the formula that does not require knowing n .

$$\begin{aligned} S_n &= \frac{a_1 - a_n r}{1 - r} \\ &= \frac{3 - 192 \cdot (-2)}{1 - (-2)} \\ &= \frac{3 + 384}{3} \\ &= \frac{387}{3} \\ &= 129 \end{aligned}$$

The sum of the series is 129.

B $2 + 1 + 0.5 + 0.25 + 0.125 + \dots + 0.015625$

SOLUTION

Here, $a_1 = 2$, $r = 0.5$, and $a_n = 0.015625$. Use the formula that does not require knowing n .

$$\begin{aligned} S_n &= \frac{a_1 - a_n r}{1 - r} \\ &= \frac{2 - 0.015625 \cdot 0.5}{1 - 0.5} \\ &= \frac{2 - 0.0078125}{0.5} \\ &= \frac{1.9921875}{0.5} \\ &= 3.984375 \end{aligned}$$

The sum of the series is 3.984375. ■